Irkutsk National Research Technical University
Baikal school of BRICS

09.04.02 Information systems and technology
Program: Information technologies, networks and big data

Course: Data analysis

Atalyan Alina Valerievna
alinaa@mail.ru

Practical class 2

Statistical Analysis

- Probability Distributions

Probability Distributions

Distribution	Type	Parameters	When to Use
Bernoulli	Discrete	p	Single yes/no trial.
Binomial	Discrete	n, p	Count successes in n trials.
Poisson	Discrete	λ	Rare event counts.
Uniform	Continuous	a, b	Equal probability over range.
Normal	Continuous	μ, σ	Natural phenomena measurements.
Exponential	Continuous	λ	Time between independent events.

Cleff, T. (2025). Random Variables and Probability Distributions. In: Applied Statistics and Multivariate Data Analysis for Business and Economics. Springer Texts in Business and Economics. Springer, Cham. https://doi.org/10.1007/978-3-031-78070-7_7

Probability Distributions

Distribution	Type	Parameters	When to Use
Log-Normal	Continuous	μ – mean of log (scale), σ – standard deviation of log (shape, σ > 0)	When data are positively skewed, bounded below by zero (≥ 0), and the logarithm of the variable follows a normal distribution. Common in fields like finance (stock prices), environmental science (pollutant concentrations), reliability engineering (failure times), and biology (body sizes).

Result of «Missing Data» task

Name: “factor_data.csv“ ,”imputed_data.csv”
Source: Internal company database
Size: 1148 obs. of 6 variables; 1148 obs. of 26 variables
Format: CSV
Columns:

```
> str(factor_df)
'data.frame':   1148 obs. of  6 variables:
 $ record_id  : int  1 2 3 4 6 8 9 10 12 13 ...
 $ outcome    : int  0 0 0 0 0 1 0 1 1 0 ...
 $ factor_eth : int  1 1 1 1 1 1 1 1 1 2 ...
 $ factor_h   : int  0 0 0 0 0 0 0 1 0 1 ...
 $ factor_pcos: int  NA NA NA NA NA NA NA NA NA 1 ...
```

```
> str(imputed_df)
'data.frame':   1148 obs. of  26 variables:
 $ record_id          : int  1 2 3 4 6 8 9 10 12 13 ...
 $ hormone1           : num  0.5 1.9 1.2 3.8 1.6 1.6 0.42 2.7 1 1.8 ...
 $ hormone2           : int  314 196 311 409 418 309 415 450 198 569 ...
 $ hormone3           : num  9.9 6.8 1.2 3.2 3.8 2.7 5.9 7 9 20.6 ...
 $ hormone4           : num  4.2 6.8 3.2 2.7 7.2 1.6 9.1 4.5 5.2 8.3 ...
 $ hormone5           : num  322.7 190.9 153.7 225.7 99.9 ...
 $ hormone6           : num  166.1 41 140.8 93.4 24.7 ...
 $ hormone7           : num  5.7 6.8 5.4 12.7 1.6 9.9 7.5 4 4.2 1.7 ...
 $ hormone8           : num  129 260 81.1 148 121 120 135 199 149 238 ...
```

Data Sets

Name: “data_for_analysis“ (from data frames factor_df and imputed_handle_MD_df_final q.v. practice 1)
Source: Internal company database
Size: 1148 obs. of 31 variables
Format: CSV
Columns:

record_id (int) Unique order identifier
outcome (int) Presence of a tumor
hormone1 (num) result of hormone 1 level assessment
…
hormone4 (num) result of hormone 4 level assessment
h_index_34 (num) index (hormonal ratio)
…
…
antioxidant5 (num) result of antioxidant 5 level assessment
factor_eth (int)) Ethnicity
factor_h (int) Increased hormones
factor_pcos (int) Diagnosis
factor_prl (int) Increased hormone

Potential Use for the Practical class 3:

select the most optimal distribution law and evaluate its parameters;
creating a table with descriptive statistics for a data set;

repeat

Missing Data

```
> skim(df)
── Data Summary ────────────────────────
                           Values
Name                       df
Number of rows             1148
Number of columns          41
_______________________
Column type frequency:
  numeric                  41
_______________________
Group variables            None
```

repeat

Missing Data

— Variable type: numeric ——————————————————————————————————————

	skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
1	record_id	0	1	1306.	746.	1	655	1258	1946.	2694	
2	outcome	1	0.999	0.139	0.347	0	0	0	0	1	
3	hormone1	14	0.988	1.81	1.51	0.05	1.1	1.5	2.1	26.1	
4	hormone2	16	0.986	382.	259.	31	228.	320	458	2798	
5	hormone3	19	0.983	9.12	11.7	0	3.5	5.7	9.4	118.	
6	hormone4	17	0.985	7.80	11.4	0.1	3.8	5.5	7.4	131.	
7	h_index_34	19	0.983	1.34	1.13	0	0.674	1.02	1.67	18	
8	hormone5	18	0.984	327.	323.	0	178.	270.	378.	5687.	
9	hormone6	16	0.986	78.7	53.5	0.76	40.7	65	101.	375.	
10	h_index_56	20	0.983	2.25	4.48	0	0.792	1.39	2.52	120.	
11	hormone7	16	0.986	5.64	3.92	0	2.68	5.1	7.6	47.1	
12	hormone8	19	0.983	176.	85.8	16.3	116	162	218	558	
13	hormone9	1009	0.121	23.8	29.1	0.1	5.75	10.9	30.4	172.	
14	hormone10_1	1078	0.0610	0.368	0.186	0.178	0.250	0.280	0.437	1	
15	hormone10_2	632	0.449	0.984	11.5	0.077	0.278	0.44	0.613	261	
16	hormone10_generated	393	0.658	0.843	9.49	0.077	0.284	0.432	0.632	261	
17	hormone11	987	0.140	12.3	4.68	1.1	9.7	13.1	16	19.2	
18	hormone12	987	0.140	13.1	3.97	0.5	10.7	14.2	16.4	19	
19	hormone13	954	0.169	441.	320.	68.9	196.	374.	535	1901	
20	hormone14	931	0.189	52.1	57.0	3.2	17.2	24.2	71.9	332	
36	antioxidant4	42	0.963	7.86	3.17	0	5.45	7.34	9.68	25	
37	antioxidant5	40	0.965	0.690	0.289	0.23	0.48	0.62	0.84	2.04	
38	factor_eth	1	0.999	1.46	0.659	1	1	1	2	3	
39	factor_h	14	0.988	0.194	0.396	0	0	0	0	1	
40	factor_pcos	983	0.144	2.6	1.20	1	1	3	4	4	
41	factor_prl	15	0.987	0.0671	0.250	0	0	0	0	1	

Probability Distributions

The choice of the most optimal distribution law is based on the maximum likelihood method and the Bayesian information criterion

The **Bayesian Information Criterion (BIC)** is a model selection metric that balances **model fit** and **complexity** (similar to AIC but with a stronger penalty for extra parameters). It is defined as:

$BIC = -2 \cdot \log(L) + k \cdot \log(n)$ $BIC = -2 \cdot \log(L) + k \cdot \log(n)$

L = Likelihood of the model

k = Number of parameters

n = Sample size

Lower BIC values indicate better models.

Recommended Additional Components

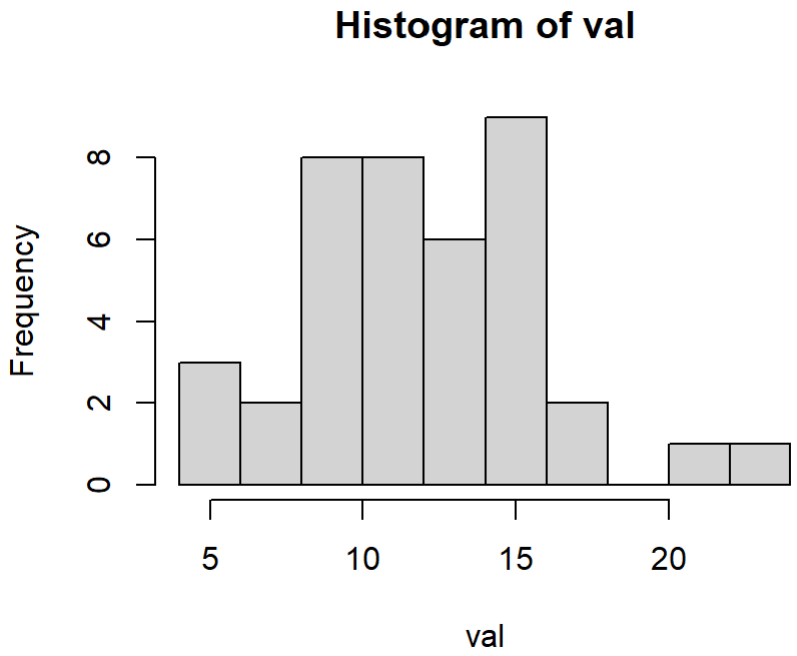
Recommended Additional Components
Essential R packages
For practical class 2:

library(MASS)
For fitdistr(val, densfun="normal") - function from the MASS package fits a normal (Gaussian) distribution to your data vector val using **Maximum Likelihood Estimation (MLE)**

Probability Distributions

For _example_distribution.csv

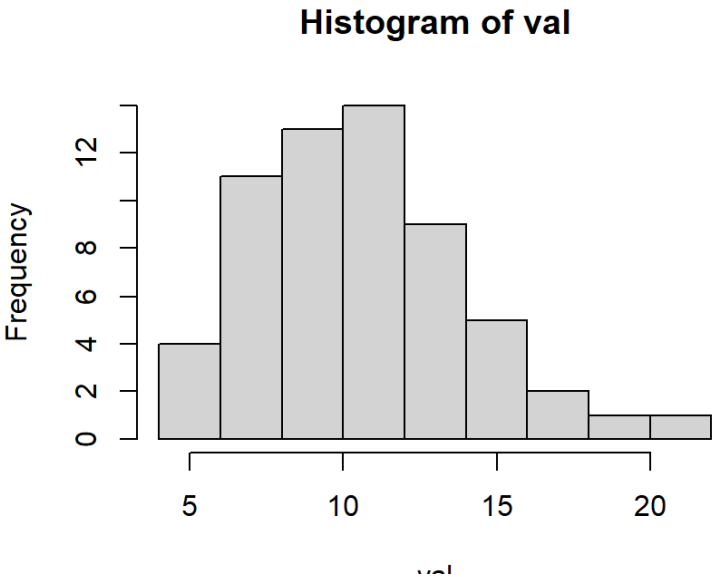
```
> summary(example_df)
 distribution           value
 Length:188         Min.   : 0.1817
 Class :character   1st Qu.: 8.0000
 Mode  :character   Median :11.0000
                    Mean   :11.5320
                    3rd Qu.:14.0000
                    Max.   :51.9546
> #building histograms for example
> # normal distribution
> val<-example_df[example_df$distribution=="norm",]$value
>
> mean(val)
[1] 12.09659
>
> sd(val)
[1] 4.024829
>
> hist(val)
>
> fit<-fitdistr(val, densfun="normal")
>
> fit
      mean          sd
  12.0965929    3.9742006
 ( 0.6283763) ( 0.4443291)
```



Probability Distributions

For example_distribution.

```
> #lognormal distribution
> val<-example_df[example_df$distribution=="lognorm",]$value
>
> mean(val)
[1] 10.61198
>
> sd(val)
[1] 3.333678
>
> hist(val)
>
> fit<-fitdistr(val, densfun="lognormal")
>
> fit
    meanlog       sdlog
  2.31325613   0.31472528
 (0.04063086) (0.02873036)
```




```
> fit<-fitdistr(val, densfun="lognormal")
>
> fit
    meanlog       sdlog
  2.31325613   0.31472528
 (0.04063086) (0.02873036)
```

```
> unname(fit$estimate[1])
[1] 2.313256
>
> unname(fit$estimate[2])
[1] 0.3147253
```

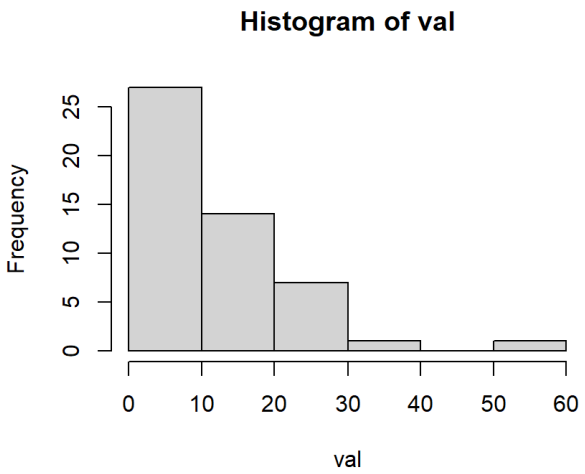
```
> m_log<-exp(unname(fit$estimate[1]))*sqrt(exp(unname(fit$estimate[2])^2))
> m_log
[1] 10.62046
> sd_log<-sqrt(exp(2*unname(fit$estimate[1]))*(exp(unname(fit$estimate[2])^2)-1)*sqrt(exp(unname
(fit$estimate[2])^2)))
> sd_log
[1] 3.34321
```

Probability Distributions

For example_distribution.csv

```
> #exponential distribution
>
> val<-example_df[example_df$distribution=="exp",]$value
>
> mean(val)
[1] 12.00853
>
> sd(val)
[1] 10.41256
>
> hist(val)
```

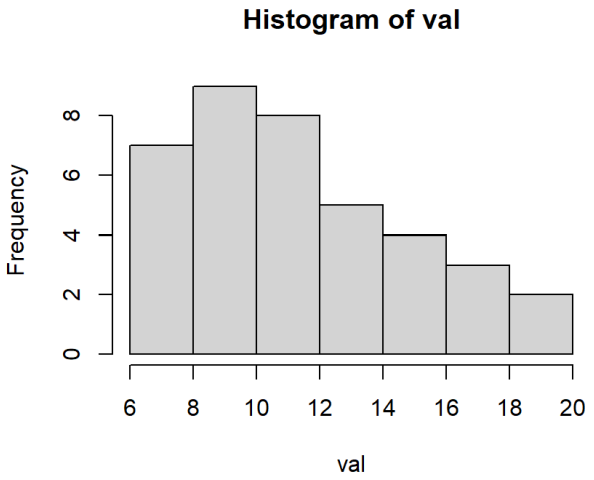
```
> fit<-fitdistr(val, densfun="exponential")
>
> fit
      rate
  0.08327416
 (0.01177674)
>
> unname(fit$estimate[1])
[1] 0.08327416
>
> m_exp<-1/unname(fit$estimate[1])
> m_exp
[1] 12.00853
```



Probability Distributions

For example distribution.csv

```
> #Poisson distribution
> val<-example_df[example_df$distribution=="pois",]$value
>
> mean(val)
[1] 11.76316
>
> sd(val)
[1] 3.420402
>
> hist(val)
>
> fit<-fitdistr(val, densfun="Poisson")
>
> fit
     lambda
  11.7631579
 ( 0.5563783)
>
> unname(fit$estimate[1])
[1] 11.76316
>
> sd_pois<-sqrt(unname(fit$estimate[1]))
> sd_pois
[1] 3.429746
```



Probability Distributions

For example_distribution.csv

```
> #Selecting a Distribution Model
>
> val<-example_df[example_df$distribution=="lognorm",]$value
>
> fit_1<-fitdistr(val, densfun="normal")
> fit_2<-fitdistr(val, densfun="lognormal")
> fit_3<-fitdistr(val, densfun="exponential")
> #Bayesian Information Criterion calculation
> BIC(fit_3)
[1] 407.5324
> #calculation of the Bayesian information criterion for all models
> BIC_value<-c(BIC(fit_1), BIC(fit_2), BIC(fit_3))
>
> #forming a vector with the name of the models
> distribution<-c("normal", "lognormal", "exponential")
>
> #combining the results into a final table
> rez<-data.frame(BIC_value=BIC_value, distribution=distribution)
>
> #sort table in ascending order of Bayesian Information Criterion value
> rez<-rez[order(rez$BIC_value, decreasing=F),]
>
> rez
  BIC_value distribution
2  317.3254    lognormal
1  321.9420       normal
3  407.5324  exponential
```

Probability Distributions

```
> #calculation of absolute values of the confidence interval for the mean of a lognormal distribu
tion
> error_min<-unname(fit_2$estimate[1])-unname(fit_2$sd[1])
> error_max<-unname(fit_2$estimate[1])+unname(fit_2$sd[1])
>
> error_min
[1] 2.272625
> error_max
[1] 2.353887
>
> m<-exp(unname(fit_2$estimate[1]))*sqrt(exp(unname(fit_2$estimate[2])^2))
> value_error_min<-exp(error_min)*sqrt(exp(unname(fit_2$estimate[2])^2))
> value_error_max<-exp(error_max)*sqrt(exp(unname(fit_2$estimate[2])^2))
>
> value_error_min
[1] 10.19759
> m
[1] 10.62046
> value_error_max
[1] 11.06086
```

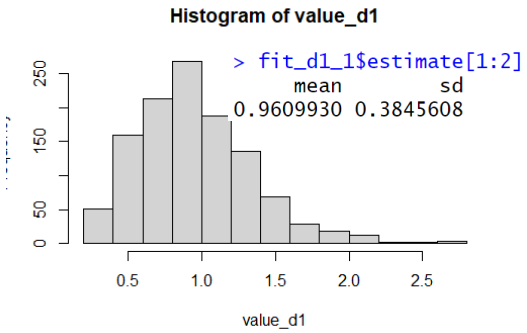
Probability Distributions

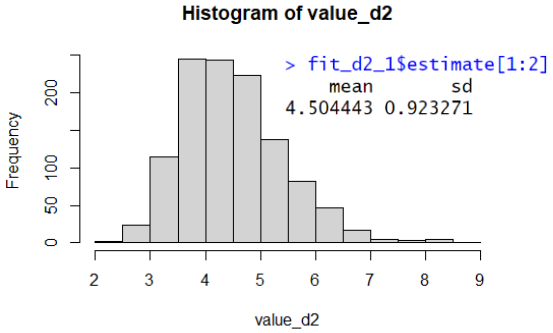
```
$ lipids1              : num  0.4 0.54 0.45 0.38 0.48 0.34 0.49 0.62 1.1 0.51 ...
$ lipids2              : num  5.05 3.9 3.5 4.84 3.96 3.85 3.93 4.3 3.89 3.72 ...
$ lipids3              : num  1.23 1.67 1.29 1.32 1.85 1.42 1.26 1.3 1.11 0.78 ...
$ lipids4              : num  3.64 1.98 2.01 3.35 1.89 2.28 2.45 2.72 2.28 2.28
```

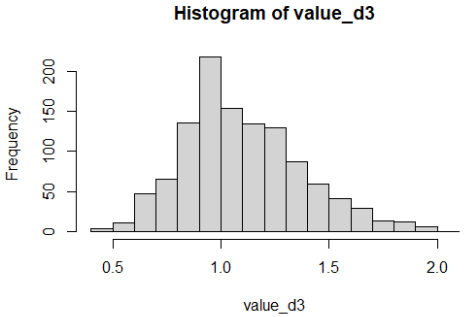
Distributions =?

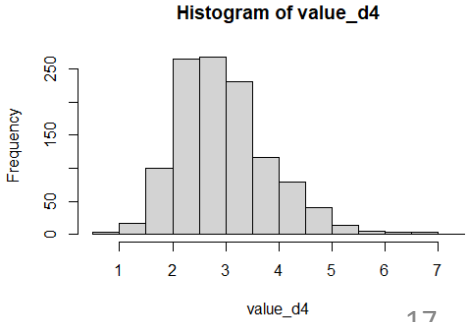
```
> result_d1
  BIC_value_d1 distribution
1    1077.7942       normal
2     953.3272    lognormal
3    2211.6922  exponential
```

```
> result_d2
  BIC_value_d2 distribution
1     3088.679       normal
2     2981.184    lognormal
3     5758.673  exponential
```









Task 1

Deadline for Subgroup 1 13.04.2026 23:59
Deadline for Subgroup 1 20.04.2026 23:59

Dataset “data_for_analysis” (except lipids 5)

1. estimate the distribution of continuous variables in data by group (outcome)
2. create a table with descriptive statistics and specify the parameters according to the selected distribution (by group)

for extra points!

1. find the error in the data and fix it (missing data «lipids 5»)
2. estimate the distribution of continuous variables in data by groups (outcome)
3. create a table with descriptive statistics and specify the parameters according to the selected distribution (by groups)